

W 146 x L 240 mm
Both side print

Camlo[®]

Amlodipine Besilate

COMPOSITION

Camlo[®] 5 tablet: Each tablet contains 5 mg of Amlodipine as besilate salt.

Camlo[®] 10 tablet: Each tablet contains 10 mg of Amlodipine as besilate salt.

PRODUCT DESCRIPTION

Camlo[®] 5 tablet: A white colored round tablet plain on both sides.
Camlo[®] 10 tablet: A white colored round tablet, embossed SQUARE on one side and break line on other side.

PHARMACODYNAMICS

Pharmacotherapeutic group: Dihydropyridine derivatives
ATC code: C08C A01

Amlodipine is a calcium antagonist and inhibits the influx of calcium ions into cardiac and smooth muscle cells. The mechanism of the antihypertensive action is due to the direct spasmolytic effect on vascular smooth muscle cells. The precise mechanism by which amlodipine relieves angina pectoris has not been fully determined, but the following two actions play a role:

1. Amlodipine dilates peripheral arterioles and thus reduces the peripheral resistance (afterload) against which the heart pumps. This unloading of the heart reduces myocardial energy consumption and oxygen requirements.

2. Dilatation of the main coronary arteries and the coronary arterioles also probably plays a role in its action. This dilation increases the supply of oxygen to myocardial muscle in patients with Prinzmetal anginal attack.

In *patients with hypertension*, once daily dosing provides clinically significant reductions of blood pressure (in both supine and standing positions) that persist for 24 hours.

In *patients with angina pectoris*, once daily administration of amlodipine increases total exercise time, the delay of occurrence of anginal attack and the delay of the occurrence of a 1-mm ST interval. Amlodipine decreases both attack frequency and glyceryl trinitrate tablet consumption.

In haemodynamic studies in *patients with heart failure* and in clinical studies based on exercise tests in patients with NYHA class II-IV cardiac decompensation, amlodipine was found not to cause any clinical deterioration, as measured by exercise tolerance, left ventricular ejection fraction and clinical signs and symptoms.

In a placebo-controlled study (PRAISE) designed to evaluate patients with NYHA class III-IV cardiac decompensation treated with digoxin, diuretics and ACE inhibitors, amlodipine was shown not to cause any increase in the risk of death or in the combined risk of mortality and morbidity in patients with cardiac decompensation.

Patients with cardiac failure should be treated with caution. In a long-term study including patients suffering from severe heart failure (NYHA grade III and IV) the reported incidence of pulmonary oedema was higher in the amlodipine treated group than in the placebo group, but this was not indicating an aggravation of the heart failure.

PHARMACOKINETICS

Absorption/Distribution

After oral administration of therapeutic doses amlodipine is slowly absorbed from the gastrointestinal tract. The absorption of amlodipine is unaffected by the concomitant intake of food. The absolute bioavailability of the unchanged compound is estimated as 64-80%. Peak plasma levels are reached 6 to 12 hours post-dose. The volume of distribution is about 20 l/kg. The pKa of amlodipine is 8.6. Plasma protein binding in vitro is approximately 98%.

Metabolism/Elimination

The plasma elimination half-life is about 35 to 50 hours. Steady state plasma levels are reached after 7-8 consecutive days. Amlodipine is predominantly metabolised to inactive metabolites. About 60% of the administered dose is excreted in the urine, about 10% of which is in the form of unchanged amlodipine.

In the elderly

The time to reach peak plasma concentrations is the same in elderly and younger patients. Clearance may be reduced in elderly patients so that the area under the curve (AUC) and the terminal elimination

half-life are increased. The recommended dosage regimen for elderly patients is however the same, although caution should be exercised when increasing the dosage.

In patients with impaired renal function

Amlodipine is extensively biotransformed to inactive metabolites. Ten percent of the substance is excreted unchanged in the urine. Changes in amlodipine plasma concentration are not correlated with the degree of renal impairment. In these patients amlodipine may be administered at the normal dosage. Amlodipine is not dialysable.

Patients with hepatic impairment:

The half-life of amlodipine is prolonged in patients with impaired hepatic function.

INDICATION

Amlodipine is indicated for the first line treatment of hypertension and can be used as the sole agent to control blood pressure in the majority of patients. Patients not adequately controlled on a single antihypertensive agent may benefit from the addition of amlodipine, which has been used in combination with a thiazide diuretic, alpha blockers, beta adrenoceptor blocking agent, or an angiotensin-converting enzyme inhibitor.

Amlodipine is indicated for the first line treatment of myocardial ischemia, whether due to fixed obstruction (stable angina) and/or vasospasm/vasoconstriction (Prinzmetal's or variant angina) of coronary vasculature. Amlodipine may be used where the clinical presentation suggests a possible vasospastic/ vasoconstrictive component but where vasospasm/vasoconstriction has not been confirmed. Amlodipine may be used alone, as monotherapy, or in combination with other antianginal drugs in patients with angina that is refractory to nitrates and/or adequate doses of beta blockers.

RECOMMENDED DOSAGE

For both hypertension and angina the usual initial dose is 5 mg amlodipine once daily which may be increased to a maximum dose of 10mg depending on the individual patient's response.

No dose adjustment of amlodipine is required upon concomitant administration of thiazide diuretics, beta-blockers, and angiotensin-converting enzyme inhibitors.

Use in the elderly

Normal dosage regimens are recommended. Amlodipine, used at similar doses in elderly or younger patients, is equally well tolerated.

Use in children

Safety and effectiveness of amlodipine in children have not been established.

Use in patients with impaired hepatic function

As with all calcium antagonists, amlodipine half-life is prolonged in patients with impaired liver function and dosage recommendations have not been established. The drug should therefore be administered with caution in these patients.

Use in renal failure

Amlodipine may be used in such patients at normal doses. Changes in amlodipine plasma concentrations are not correlated with degree of renal impairment. Amlodipine is not dialyzable.

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATION

Amlodipine is contraindicated in patients with known hypersensitivity to dihydropyridine (e.g. amlodipine, nifedipine, nicardipine, isradipine).

WARNING AND PRECAUTION

Amlodipine should be administered with caution to patients with low cardiac reserve.

Patients with heart failure

Patients with cardiac failure should be treated with caution. In a long-term study including patients suffering from severe heart failure (NYHA grade III and IV) the reported incidence of pulmonary oedema was higher in the amlodipine treated group than in the placebo group, but this was not indicating an aggravation of the heart failure.

Use in patients with impaired hepatic function

The half-life of amlodipine is prolonged in patients with impaired liver function; dosage recommendations have not been established. Amlodipine should therefore be administered with caution in these patients.

Use in elderly patients

In the elderly, increase of the dosage should take place with care.

Use in children

Amlodipine should not be given to children due to insufficient clinical experience.

Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

INTERACTIONS WITH OTHER MEDICAMENTS

Effects of other medicinal products on amlodipine

CYP3A4 inhibitors: A study of elderly patients has shown that diltiazem inhibits metabolism of amlodipine, probably via CYP3A4, since plasma concentration increases by approx. 50% and the effect of amlodipine is increased. It cannot be excluded that stronger inhibitors of CYP3A4 (i.e. ketoconazole, itraconazole, ritonavir) increase the plasma concentration of amlodipine to a greater extent than diltiazem. Caution should be exercised in combination of amlodipine and CYP3A4 inhibitors.

CYP3A4 inducers: There is no information available on the effect of CYP3A4 inducers (i.e. rifampicin, St. John's wort) on amlodipine. Co-administration may lead to reduced plasma concentration of amlodipine. Caution should be exercised in combination of amlodipine and CYP3A4 inducers.

In clinical interaction studies grapefruit juice, cimetidine, aluminium/magnesium (antacid) and sildenafil did not affect the pharmacokinetics of amlodipine.

Effects of amlodipine on other medicinal products

Amlodipine may potentiate the effect of other anti-hypertensives as beta-adrenoceptor blocking agents, ACE-inhibitors, alpha-1-blockers and diuretics. In patients with an increased risk (for example after myocardial infarction) the combination of a calcium channel blocker with a beta-adrenoceptor blocking agent may lead to heart failure, to hypotension and to a (new) myocardial infarction.

In clinical interaction studies, amlodipine did not affect the pharmacokinetics of atorvastatin, digoxin, warfarin or ciclosporin.

There is no effect of amlodipine on laboratory testing.

PREGNANCY AND LACTATION

There are no adequate data from the use of amlodipine in pregnant women.

In animal studies effects on reproduction were found at high dosages. The potential risk for humans is unknown. Accordingly amlodipine should not be used during pregnancy unless clearly needed.

It is not known whether amlodipine is excreted in breast milk. It is advised to stop breastfeeding during treatment with amlodipine.

SIDE EFFECTS

Very common: >1/10

Common: >1/100 and <1/10

Uncommon: >1/1000 and <1/100

Rare: >1/10 000 and <1/1000

Very rare: <1/10 000 including isolated cases

Blood and lymphatic system disorders:

Very rare: Leukocytopenia, thrombocytopenia.

Endocrine disorders:

Uncommon: Gynaecomastia.

Metabolism and nutrition disorders:

Very rare: Hyperglycaemia.

Nervous system disorders:

Common: Headache (especially in the beginning of the treatment), fatigue, dizziness, asthenia

Uncommon: Malaise, dry mouth, tremor, paraesthesia, increased sweating

Rare: Taste changes

Very rare: Peripheral neuropathy

Eye disorders:

Uncommon: Visual disturbances.

Psychiatric disorders:

Uncommon: Sleep disorder, irritability, depression

Rare: Confusion, mood changes including anxiety.

Ear and labyrinth disorders:

Rare: Tinnitus

Cardiac disorders:

Common: Palpitations

Uncommon: Syncope, tachycardia, chest pain, at the beginning of treatment aggravation of angina pectoris may happen, isolated cases of myocardial infarction and arrhythmias (including extrasystole, ventricular tachycardia, bradycardia and atrial arrhythmias) and chest pain have been reported in patients with coronary artery disease, but a clear association with amlodipine has not been established

Vascular disorders:

Uncommon: Hypotension

Very rare: Vasculitis.

Respiratory, thoracic and mediastinal disorders:

Uncommon: Dyspnoea, rhinitis

Very rare: Cough.

Gastrointestinal disorders:

Common: Nausea, dyspepsia, abdominal pain

Uncommon: Vomiting, diarrhoea, constipation, gingival hyperplasia

Very rare: Gastritis.

Hepato-biliary disorders:

Rare: Elevated liver enzymes, jaundice, hepatitis

Very rare: Pancreatitis

Skin and subcutaneous tissue disorders:

Very common: Ankle swelling

Common: Facial flushing with heat sensation, especially at the beginning of the treatment

Uncommon: Exanthema, pruritus, urticaria, alopecia, skin discolouration

Very rare: Angioedema, isolated cases of allergic reactions including pruritus, rash, angioedema and erythema exsudativum multiforme, exfoliative dermatitis and Stevens Johnson syndrome and Quincke oedema have been reported.

Musculoskeletal, connective tissue and bone disorders:

Uncommon: Muscle cramps, back pain, myalgia and arthralgia.

Renal and urinary disorders:

Uncommon: Increased micturition frequency.

Reproductive system and breast disorders:

Uncommon: Impotence.

General disorders and administration site conditions:

Uncommon: Increase or decrease of weight.

SYMPTOMS AND TREATMENT OF OVERDOSE

In humans, experience with intentional overdose is limited. Available data suggest that overdose >100 mg) could result in excessive peripheral vasodilatation with subsequent marked and probably prolonged systemic hypotension.

Clinically significant hypotension due to amlodipine overdose calls for active cardiovascular support including frequent monitoring of cardiac and respiratory function, elevation of extremities, and attention to circulating fluid volume and urine output.

A vasoconstrictor may be helpful in restoring vascular tone and blood pressure, provided that there is no contraindication to its use. Intravenous calcium gluconate may be beneficial in reversing the effects of calcium channel blockade. Gastric lavage may be worthwhile in some cases. In healthy volunteers the use of charcoal up to 2h after administration of amlodipine 10 mg has been shown to reduce the absorption rate of amlodipine. Since amlodipine is highly protein-bound, dialysis is not likely to be of benefit.

STORAGE CONDITION

Store at temperature not exceeding 30°C.

Keep medicine out of reach of children.

Jauhkan daripada kanak-kanak.

SHELF LIFE

3 years

SUPPLY

Camlodin 5 Tablet: Box containing 2x15 / 4x15 tablets in blister pack.

Camlodin 10 Tablet: Box containing 2x15 / 4x15 tablets in blister pack.

MANUFACTURER

SQUARE PHARMACEUTICALS LTD.

Dhaka Unit, Kaliakoir, Gazipur, Bangladesh

NAME & ADDRESS OF MARKETING

AUTHORIZATION HOLDER:

Medispec (M) Sdn Bhd

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DECEMBER 2017

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